

On Σ_3^1 and Σ_4^1 -uniformization

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Abstract

We construct a model of set theory in which the lightface Σ_3^1 -uniformization property holds, yet the lightface Σ_4^1 -uniformization property fails, separating these two principles for the first time.

1 Introduction

The uniformization problem—asking whether every relation in a given pointclass contains a function with the same domain in the same pointclass—has been a central driving force in descriptive set theory since its inception. While the Axiom of Choice guarantees the existence of uniformizing functions in the abstract, the definability of such functions is a much more delicate matter.

At the foundational levels of the projective hierarchy, the situation is entirely resolved within standard set theory. A classical result of Kondo [3] establishes that every Π_1^1 relation can be uniformized by a Π_1^1 function. As a direct consequence of projecting this result, the Σ_2^1 -uniformization property also holds in ZFC.

However, for pointclasses beyond the second level of the projective hierarchy, ZFC alone is insufficient to decide uniformization. The behavior of uniformization at the third level and above heavily depends on the underlying structural properties of the universe, historically dividing into two mutually exclusive paradigms: the existence of definable well-orders and the assumption of strong determinacy hypotheses.

In canonical inner models, such as Gödel's constructible universe L or larger core models, the reals admit a good projective well-ordering. For instance, in L , the canonical well-ordering of the reals is Δ_2^1 , which systematically implies that the Σ_n^1 -uniformization property holds for all $n \geq 2$. More generally, if a model admits a Δ_n^1 good well-ordering of the reals, the standard structural consequences force Σ_k^1 -uniformization to be true for all $k \geq n$. Thus, in all prior known models where the Σ_3^1 -uniformization property was established, it was achieved via the presence of a projective well-order (or a similarly rigid structural property) that inextricably carried the Σ_4^1 -uniformization property along with it.

Conversely, under Projective Determinacy (PD), the picture is radically different. The periodicity theorems of Martin, Moschovakis, and Solovay [4] dictate that uniformization alternates: at the third level, it is the Π_3^1 relations that uniformize, while Σ_3^1 -uniformization explicitly fails. At the fourth level, Σ_4^1 -uniformization holds.

This historical dichotomy has left open a natural structural question regarding the independence of these projective levels from one another. Specifically, must the truth of Σ_3^1 -uniformization always drag Σ_4^1 -uniformization with it?

In this paper, we answer this question in the negative. We provide a novel construction that breaks the classical reliance on a good projective well-ordering of the reals. Our main theorem demonstrates that it is possible to surgically preserve uniformization at the third level while arranging for its failure at the fourth.

Theorem 1.1. *There exists a model of set theory in which the lightface Σ_3^1 -uniformization property holds, yet the lightface Σ_4^1 -uniformization property fails.*

By doing so, we show that the Σ_4^1 -uniformization property does not strictly follow from the Σ_3^1 -uniformization property, effectively separating these two properties at the projective levels and providing a new structural paradigm distinct from both standard inner models and determinacy.

2 The Ground Model and the Coding Machinery

To establish our separation result, we rely heavily on the foundational coding machinery developed in [2]. In this section, we provide a detailed, rigorous exposition of the ground model W , the coding mechanism via independent Suslin trees, and the base framework of allowable forcings.

2.1 Independent Suslin Trees and Almost Disjoint Coding

Our construction relies on the ability to generically destroy Suslin trees independently of one another.

Definition 2.1. Let $\vec{T} = (T_\alpha \mid \alpha < \kappa)$ be a sequence of Suslin trees. We say that the sequence is an *independent family of Suslin trees* if for every finite set of pairwise distinct indices $e = \{e_0, e_1, \dots, e_n\} \subset \kappa$, the product $T_{e_0} \times T_{e_1} \times \dots \times T_{e_n}$ is a Suslin tree.

In L , we can canonically define an ω_1 -sequence of such trees.

Theorem 2.2. *Assume that $\aleph_1 = \aleph_1^L$ and that $(M, \in)$ is a transitive, ω_1 -containing, uncountable model of $\text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$. Then there is an independent sequence $\vec{T} = (T_\alpha \mid \alpha < \omega_1)$ of L -Suslin trees, and the sequence $\vec{T}$ is uniformly $\Sigma_1(\{\omega_1\})$ -definable over M . Specifically, there is a Σ_1 -formula ϕ with ω_1 as the unique parameter such that the relation $\{(t, \gamma, \eta) \mid \gamma, \eta < \omega_1 \wedge t \in T_\eta^\gamma\}$ is definable over M using ϕ , where T_η^γ denotes the γ -th level of T_η .*

We split $\vec{T}$ into two independent sequences, which will be utilized to separate our projective properties:

$$\vec{S}^0 := \{S_\alpha \mid \alpha \text{ is even}\} \quad \text{and} \quad \vec{S}^1 := \{S_\alpha \mid \alpha \text{ is odd}\}.$$

To encode these branches into reals, we employ the almost disjoint coding forcing $\mathbb{A}_D(X)$. We fix a canonical L -definable almost disjoint family $D = \{d_\alpha \mid \alpha < \aleph_1\} \subset [\omega]^\omega$. For a set $X \subset \omega_1$, $\mathbb{A}_D(X)$ is the ccc forcing that adds a real x such that $\alpha \in X$ if and only if $x \cap d_\alpha$ is finite. As $\mathbb{A}_D(X)$ has the Knaster property, it preserves the Suslinity of any Suslin tree.

2.2 The Ground Model and the Coding Forcing

We first force over L with a finite support product $\mathbb{P}^0 := \prod_{\alpha < \omega_1} S_\alpha$ to generically add an ω_1 -branch to every tree in $\vec{S} = \vec{S}^0 \cup \vec{S}^1$. Let G^0 be generic for $\mathbb{P}^0$. Next, we add ω_1 -many Cohen subsets of ω_1 using a countably supported product evaluated in L : $\mathbb{P}^1 := (\prod_{\alpha < \omega_1} \mathbb{C}(\omega_1))^L$. Let G^1 be generic for $\mathbb{P}^1$.

Our ground model is defined as:

$$W := L[G^0][G^1].$$

Since $\mathbb{P}^1$ is σ -closed over L , $\aleph_1$ is preserved and $W \models \text{CH}$.

Let $x \in W$ be a real, $m, k \in \omega$, and let $l \in \{0, 1\}$ and $\eta < \omega_1$. We define a localized forcing $\text{Code}(x, l, \eta)$ over W which codes the tuple x into the branches of $\vec{S}^l$. We illustrate this for $l = 1$.

We denote the η -th coordinate of the generic filter G^1 by $g_\eta \subset \omega_1$. Fixing a canonically constructible bijection $\rho : ([\omega_1]^\omega)^L \rightarrow \omega_1$, we define a subset $h \subset \omega_1$:

$$h := \{\rho(g_\eta \cap \alpha) \mid \alpha < \omega_1\}.$$

This h indexes the ω -blocks of $\vec{S}^1$ where x will be coded. We define the block assignment $A \subset \omega_1$ as:

$$A := \{\omega\gamma + 2n \mid \gamma \in h, n \notin x\} \cup \{\omega\gamma + 2n + 1 \mid \gamma \in h, n \in x\}.$$

Let $X \subset \omega_1$ canonically code the set A and the corresponding set of generic branches $\{b_\beta \subset S_\beta^1 \mid \beta \in A\}$. Working in $L[X]$, one can decode x by inspecting the ω -block of trees starting at any $\gamma \in h$:

(*) (γ, x) : $n \in x$ iff $S_{\omega\gamma+2n+1}^1$ has an ω_1 -branch, and $n \notin x$ iff $S_{\omega\gamma+2n}^1$ has an ω_1 -branch.

Applying a variation of David's trick [1], we transform X into a set $Y \subset \omega_1$ such that Y codes a sequence of countable elementary submodels that correctly decode this branch structure. We define our coding forcing as the almost disjoint coding of this specific Y :

$$\text{Code}(x, 1, \eta) := \mathbb{A}_D(Y).$$

The fundamental effect of this forcing is the addition of a generic real r that acts as a witness to the following Π_2^1 formula in the parameters r and x :

$\Psi(r, x) :=$ For any countable, transitive model $M \models \text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$ such that $\omega_1^M = (\omega_1^L)^M$ and $r \in M$, M can construct $L[r]^M$, which in turn satisfies that there is a transitive ZFC^- -model $\bar{N}$ of size $\aleph_1^M$ such that $\bar{N} \models (*) (\gamma, x)$ for an $\aleph_1^M$ -sized set of ordinals γ .

When $\Psi(r, x)$ holds, we say that x is coded into $\bar{S}^1$. The forcing $\text{Code}(x, 0, \eta)$ is defined identically using $\bar{S}^0$.

2.3 Allowable Forcings

To iterate these coding forcings without inadvertent interference, we restrict ourselves to iterations guided by specific bookkeeping functions.

Definition 2.3. Let $\alpha < \omega_1$ and let $F \in L$, $F : \alpha \rightarrow L$ be a bookkeeping function. A finite support iteration $\mathbb{P} = (\mathbb{P}_\beta \mid \beta < \alpha)$ is called *0-allowable (relative to F)* if F determines $\mathbb{P}$ inductively as follows:

Assume $\beta \geq 0$ and $\mathbb{P}_\beta$ is defined. Let G_β be $\mathbb{P}_\beta$ -generic over W . Assume $F(\beta) = (\dot{x}, \dot{m}, \dot{k}, \dot{l}, \dot{\eta})$ evaluates to a real x , integers m, k , an index $l \in \{0, 1\}$, and an ordinal $\eta < \omega_1$.

- **The Freshness Condition:** If there is a $\gamma < \beta$ where the iteration evaluated $F(\gamma) = (\dot{a}', \dot{m}', \dot{k}', \dot{l}', \dot{\eta}')$ such that $\dot{\eta}'^{G_\gamma} = \eta$ (i.e., the coding area η has already been utilized), then $\mathbb{P}(\beta)^{G_\beta}$ is the trivial forcing.
- **Coding:** If the Freshness Condition is met (no such $\gamma < \beta$ exists), then $\mathbb{P}(\beta)^{G_\beta} := \text{Code}(x, l, \eta)$.

For a generic filter $G \subset \mathbb{P}$, the set of *coding areas* C^G is defined as the set of all $\eta < \omega_1$ successfully utilized by a non-trivial factor $\text{Code}(x, l, \eta)$ in the iteration.

This notion naturally lifts to iterations defined over extensions of W .

Definition 2.4. Let $\mathbb{P}$ be an allowable forcing over W of length δ_1 , and let $G_\mathbb{P}$ be $\mathbb{P}$ -generic over W . A finite support iteration $\mathbb{Q} = (\mathbb{Q}_\beta \mid \beta < \delta_2)$ in $W[G_\mathbb{P}]$ is called *allowable over the extension $W[G_\mathbb{P}]$* if there exists a bookkeeping function $F_\mathbb{Q} : \delta_2 \rightarrow W[G_\mathbb{P}]$ such that $\mathbb{Q}(\beta)^{G_\mathbb{Q}_\beta}$ is determined exactly as in Definition 2.3, but subject to the **Global Freshness Condition**: there is no prior stage in $\mathbb{Q}$ ($\alpha < \beta$) and no stage in the base forcing $\mathbb{P}$ ($\xi < \delta_1$) that utilized the proposed coding area η .

Lemma 2.5. Let $\mathbb{P}$ be an allowable forcing over W of length δ_1 , and suppose $\mathbb{P} \Vdash \text{“}\dot{\mathbb{Q}} \text{ is allowable over } W[G_\mathbb{P}] \text{ of length } \delta_2\text{”}$. Then the two-step iteration $\mathbb{P} * \dot{\mathbb{Q}}$ is an allowable forcing over W of length $\delta_1 + \delta_2$.

2.4 The Safety of the Iteration: No Unwanted Codes

A critical feature of allowable forcings is that they do not accidentally encode unwanted information into the sequences $\vec{S}^0$ or $\vec{S}^1$. When the iteration runs linearly, we must guarantee that the only tuples validating the coding formula Ψ are exactly those explicitly targeted by the bookkeeping function.

Lemma 2.6 (No Unwanted Codes). *Let $\mathbb{P} = (\mathbb{P}_\beta \mid \beta < \delta)$ be an allowable forcing over W , and let $G \subset \mathbb{P}$ be a generic filter over W . For $l \in \{0, 1\}$, let C^l be the set of tuples strictly targeted by a non-trivial coding factor into $\vec{S}^l$ along the iteration. Then in $W[G]$, if t is a tuple and there exists a real r witnessing $\Psi(r, t)$ for $\vec{S}^l$, then $t \in C^l$.*

Proof. We outline the proof for $l = 1$; the case for $l = 0$ is identical. We work in $W[G]$. Let $(g_\beta \mid \beta < \delta)$ be the sequence of countable coding areas utilized by $\mathbb{P}$. Via our canonical bijection ρ , these generate a family of sets $h_\beta \subset \omega_1$. By the Global Freshness Condition, all utilized coding areas g_β are distinct coordinates of the Cohen generic G^1 , ensuring the family $\{h_\beta \mid \beta < \delta\}$ is almost disjoint.

Assume for a contradiction that there is a tuple $t \notin C^1$ which satisfies $\Psi(r, t)$ for some generic real $r \in W[G]$. By the definition of the coding formula, r defines an unbounded set $h \subset \omega_1$ such that for every $\gamma \in h$, the inner model $L[r]$ correctly decodes the bits of t via the presence of ω_1 -branches in the ω -block of $\vec{S}^1$ starting at γ .

Because h is unbounded and $\{h_\beta \mid \beta < \delta\}$ is an almost disjoint family, h cannot be covered by the finite modifications of the targeted coding areas. Since t is distinct from every tuple in C^1 , there exist $\aleph_1$ -many ordinals $\alpha \in h$ (sufficiently large to be disjoint from the intersections of the h_β) and some integer n such that, without loss of generality:

$$L[r] \models "S_{\omega_\alpha + 2n + 1}^1 \text{ has an } \omega_1\text{-branch}."$$

Yet, for all $\beta < \delta$, the iteration $\mathbb{P}$ did not actively add an ω_1 -branch to this specific tree.

Because $\mathbb{P}$ is an allowable forcing, its definition depends only on the names of the reals listed by the bookkeeping function and the specific branches through $\vec{S}^1$ utilized by the coding areas. By Easton's Lemma and the ccc nature of the iteration, we can find an uncountable, co-uncountable set of indices $I \subset \omega_1$ and a countable set $J \subset \omega_1$ such that $\mathbb{P}$ is completely definable within a proper inner model $M = L[G^0 \restriction I][G^1 \restriction J]$.

We can choose our ordinal α such that the index $\xi = \omega_\alpha + 2n + 1$ does not belong to I or to any block actively modified by the h_β sets. Consequently, the tree S_ξ^1 remains a valid Suslin tree in the inner model $M[G]$.

We can factor the ground model $W = L[G^0][G^1]$ to isolate this untouched tree. Because $\vec{S}^1$ is an independent sequence, S_ξ^1 is Suslin in M , and the allowable iteration $\mathbb{P}$ has the Knaster property, preserving its Suslinity. Thus, the forcing $\mathbb{P} \times S_\xi^1$ is ccc, meaning S_ξ^1 is ω -distributive over $M[G]$.

This implies that $M[G]$ contains no new ω_1 -branches through S_ξ^1 . Since $2^\omega \cap W[G] = 2^\omega \cap M[G]$, any real $r \in W[G]$ belongs to $M[G]$. Therefore:

$$M[G] \models \neg \exists r (L[r] \models "S_\xi^1 \text{ has an } \omega_1\text{-branch}.").$$

By upward absoluteness, this must also hold in $W[G]$, directly contradicting our assumption that r witnesses the branch in $L[r]$. Thus, no such unwanted tuple t can be coded. $\square$

3 The Full ω_1 -Iteration and Diagonalization Scheme

In this section, we seamlessly weave the two strands of our construction—the Σ_3^1 -uniformization and the Σ_4^1 -diagonalization—into a single finite support iteration of length ω_1 .

We list all lightface Σ_3^1 formulas in two free variables as $\phi_0, \phi_1, \dots$. We also list all lightface Σ_4^1 formulas in two free variables as $\varphi_0, \varphi_1, \dots$, where $\varphi_m(v_0, v_1) \equiv \exists a_0 \forall a_1 \psi_m(v_0, v_1, a_0, a_1)$ and ψ_m is a Σ_3^1 formula.

We designate a specific lightface Σ_4^1 formula $\sigma(v_0, v_1)$ which will serve as our counterexample to uniformization:

$$\sigma(v_0, v_1) \equiv \exists a_0 \forall a_1 ((v_0, v_1, a_0, a_1) \text{ is not coded into } \vec{S}^1).$$

To ensure σ cannot be uniformized by any Σ_4^1 set $A_m = \{(x, y) \mid \varphi_m(x, y)\}$, we construct an allowable iteration of length ω_1 . We utilize the disjoint sequences of independent Suslin trees: $\vec{S}^0$ is strictly reserved for locking in Σ_3^1 witnesses at odd stages, and $\vec{S}^1$ is strictly reserved for the Σ_4^1 diagonalization scheme at even stages.

3.1 Weakly Allowable and Suitable Forcings

To ensure the mathematical validity of our diagonalization—specifically the squaring argument utilized later—we must relax the strict disjointness of coding areas and the strict limits on putative value creation, while preserving the integrity of our counterexample relation σ .

Definition 3.1 (Weakly Allowable Forcing). A finite support iteration of coding forcings $\mathbb{Q}$ is called *weakly allowable* if it is guided by a bookkeeping function similarly to Definition 2.4, but with two critical relaxations:

1. We completely drop the **Global Freshness Condition**. A weakly allowable forcing is permitted to reuse the same coding area η for multiple reals.
2. We drop the restriction that putative values for a pair (x, m) can be created at most once. A weakly allowable forcing is permitted to execute the Cohen product creation step multiple times for the same pair, generating multiple disjoint sets of putative values.

To manage the diagonalization process without disrupting previously secured properties, we utilize restriction sets. A restriction set E is a collection of tuples of the form $(x, y, a_0) \in (\omega^\omega)^3$. The role of E is to act as a strict blacklist: it records the putative values and challenges for which we must absolutely prevent any future coding into the $\bar{S}^1$ sequence.

Definition 3.2 (Suitable Forcing). Let M be a generic extension of W , and let $E \in M$ be a restriction set. A finite support iteration $\mathbb{Q} = (\mathbb{Q}_\xi \mid \xi < \delta)$ over M is called *suitable with respect to E* if:

1. $\mathbb{Q}$ is a weakly allowable forcing.
2. For every stage $\xi < \delta$, every $\mathbb{Q}_\xi$ -generic filter H_ξ over M , every tuple $(x, y, a_0) \in E$, every real $a_1 \in M[H_\xi]$, and every ordinal $\eta < \omega_1$, the specific coding forcing into the $\bar{S}^1$ sequence,

$$\text{Code}(x, y, a_0, a_1, 1, \eta),$$

is never utilized as the iterand $\mathbb{Q}(\xi)^{H_\xi}$.

When working within our overarching construction at stage $\beta < \omega_1$, we will have generated a specific restriction set $E_\beta \in W[G_\beta]$. We will succinctly refer to a forcing $\mathbb{Q} \in W[G_\beta]$ as being β -suitable if it is suitable with respect to E_β .

The restriction set E_β is the critical ledger for our diagonalization. By explicitly boycotting the coding of any a_1 extending a tuple $(x, y, a_0) \in E_\beta$, a β -suitable forcing mathematically guarantees that the existential statement $\exists a_0 \forall a_1 ((x, y, a_0, a_1) \text{ is not coded into } \bar{S}^1)$ remains true for the pair (x, y) in any subsequent suitable generic extension.

Lemma 3.3 (Product of Suitable Forcings). Let $\mathbb{Q}_0$ and $\mathbb{Q}_1$ be two β -suitable forcings over $W[G_\beta]$ with associated internal restriction sets $E_{\mathbb{Q}_0}$ and $E_{\mathbb{Q}_1}$. Suppose further that $\mathbb{Q}_0$ never uses a coding factor into $\bar{S}^1$ for any tuple restricted by $E_{\mathbb{Q}_1}$, and symmetrically, $\mathbb{Q}_1$ never uses a coding factor for any tuple restricted by $E_{\mathbb{Q}_0}$. Then their finite support product $\mathbb{Q}_0 \times \mathbb{Q}_1$ can be densely embedded into a weakly allowable forcing that is suitable with respect to the combined restriction set $E_\beta \cup E_{\mathbb{Q}_0} \cup E_{\mathbb{Q}_1}$.

Proof. Because weakly allowable forcings do not require the Freshness Condition and permit multiple putative value creation steps, the interleaving of the factors of $\mathbb{Q}_0$ and $\mathbb{Q}_1$ remains a valid weakly allowable iteration. If both $\mathbb{Q}_0$ and $\mathbb{Q}_1$ generate putative values for the same pair (x, m) , the product simply generates the union of these putative values. Since $\mathbb{Q}_0$ is β -suitable, it strictly satisfies the conditions of Definition 3.2 and avoids coding any tuples in $E_\beta \cup E_{\mathbb{Q}_0}$. By our non-interference hypothesis, it also avoids coding any tuple restricted by $E_{\mathbb{Q}_1}$. Symmetrically, $\mathbb{Q}_1$ avoids coding tuples in $E_\beta \cup E_{\mathbb{Q}_1} \cup E_{\mathbb{Q}_0}$. Therefore, their finite support product cleanly avoids coding any tuples in the union $E_\beta \cup E_{\mathbb{Q}_0} \cup E_{\mathbb{Q}_1}$, preserving suitability over the combined restriction set. $\square$

3.2 The Alternating Iteration Scheme

We fix a recursive partition $\langle P_m \mid m \in \omega \rangle$ of the Baire space ω^ω into ω -many pairwise disjoint sets. A real x is assigned to the integer m if and only if $x \in P_m$.

Let $F : \omega_1 \rightarrow H(\omega_1)$ be a bookkeeping function such that every tuple in $H(\omega_1)$ appears cofinally often on both the even and odd ordinals. We fix a canonical well-ordering $<_W$ of the ground model W . We proceed by induction on $\beta < \omega_1$. Let $E_0 = \emptyset$, and for limit stages λ , $E_\lambda = \bigcup_{\xi < \lambda} E_\xi$.

At each stage β , our overarching iteration $\mathbb{P}_{\omega_1}$ will be constructed as a suitable forcing. While the broader class of suitable forcings permits multiple creation steps, our specific iteration $\mathbb{P}_{\omega_1}$ will strategically gate this process to avoid trivializing the ω_1 -sequence. We evaluate $F(\beta)$ using the generic filter G_β and proceed according to the parity of β :

1. **Odd Stages (Towards Σ_3^1 -Uniformization):** If β is odd, we interpret $F(\beta)$ as a tuple $(\dot{x}, \dot{y}, \dot{m}, \dot{\eta})$. We evaluate these names to obtain x, y, m , and η . We ask if $W[G_\beta] \models \exists z \phi_m(x, z, y)$.

If such a z exists, we let z^* be the $<_W$ -least such witness. Provided the Freshness Condition holds for the coding area η , we force with $\text{Code}(x, z^*, y, m, 0, \eta)$ to encode this specific witness into the $\tilde{S}^0$ sequence. We update our restriction set $E_{\beta+1}$ to forbid any future coding of a different witness $z' \neq z^*$ for the tuple (x, y, m) . If no such z exists, or the Freshness Condition fails, we force trivially and set $E_{\beta+1} := E_\beta$.

2. **Even Stages (Towards Failure of Σ_4^1 -Uniformization):** If β is even, we interpret $F(\beta)$ as a tuple $(\dot{x}, \dot{a}_0, \dot{m}, \dot{l}, \dot{\eta})$. If $x \notin P_m$, we force trivially and set $E_{\beta+1} := E_\beta$.

If $x \in P_m$, our strategic action depends on whether putative values for the pair (x, m) have already been defined *in our main iteration*, combined with the indicator bit $l \in \{0, 1\}$:

- **Option 1 (Creation: $l = 0$):** If putative values for (x, m) have *not* yet been defined at any earlier stage of $\mathbb{P}$, we generically adjoin η -many putative values via the finite support product of Cohen forcings $\mathbb{P}(\beta)^{G_\beta} := (\prod_{i < \eta} \mathbb{C}_i) \times (\prod_{i < \eta} \mathbb{C}_i)$. Let $\{c_i(x, m) \mid i < \eta\}$ and $\{a_0^i \mid i < \eta\}$ denote the mutually generic Cohen reals added. We permanently forbid any coding involving these pairs into $\tilde{S}^1$:

$$E_{\beta+1} := E_\beta \cup \{(x, c_i(x, m), a_0^i) \mid i < \eta\}.$$

Crucially, to ensure our iteration does not run indefinitely without attempting diagonalization, if putative values for (x, m) were already defined at a previous stage, we let $\mathbb{P}(\beta)^{G_\beta}$ be the trivial forcing.

- **Option 2 (Diagonalization: $l = 1$):** If putative values *were* already defined at an earlier stage $\gamma < \beta$, we consider the given real

a_0 . We check if there exists a putative value y such that the tuple $(x, y, a_0) \in E_\beta$. If so, we ask if there is a β -suitable forcing $\mathbb{Q} \in W[G_\beta]$ that adds a real a_1 satisfying:

$$\mathbb{Q} \Vdash \neg \psi_m(x, y, a_0, \dot{a}_1).$$

If such a $\mathbb{Q}$ exists, we let $\mathbb{Q}^*$ be the $<_W$ -least such forcing, and set $\mathbb{P}(\beta)^{G_\beta} := \mathbb{Q}^*$. This actively forces (x, y) out of A_m . We set $E_{\beta+1} := E_\beta$. If no such forcing exists, we force trivially.

3.3 The “Stuck” Condition, Squaring, and the Ledger Reals

The iteration proceeds linearly unless we reach a stage where we can no longer force the pair (x, y) out of A_m . To track the collateral damage caused to our Σ_3^1 witnesses during these interruptions, we reserve a dummy index m^* which is not the Gödel number of any actual formula. This index will be used exclusively to code a sequence of ledger reals $(R_k)_{k \in \omega}$ into $\dot{S}^0$.

Definition 3.4 (The Stuck Condition). We say the iteration is *stuck* at stage β if $F(\beta)$ evaluates to $(x, a_0, m, 1, \eta)$ with $x \in P_m$, there exists a putative value y such that $(x, y, a_0) \in E_\beta$, and for every β -suitable forcing $\mathbb{Q} \in W[G_\beta]$, we have:

$$\Vdash_{\mathbb{Q}} \forall \dot{a}_1 \psi_m(x, y, a_0, \dot{a}_1).$$

If the iteration gets stuck at stage β for a putative value y , we must radically alter the ground model. Let $\gamma < \beta$ be the stage where the putative values for (x, m) were generically added via $\mathbb{P}(\gamma)$. Let k be the number of times the iteration has gotten stuck prior to this stage. The intermediate iteration factor $\mathbb{P}_{\gamma, \beta} \cong \mathbb{P}(\gamma) * \mathbb{P}_{\gamma+1, \beta}$ is a γ -suitable forcing, and it has the property that:

$$\mathbb{P}(\gamma) \Vdash \exists \dot{a}_0 (\dot{a}_0 \text{ is a } \mathbb{P}_{\gamma+1, \beta}\text{-name} \wedge \forall \mathbb{Q} \beta\text{-suitable, } \mathbb{Q} \Vdash \forall \dot{a}_1 \psi_m(x, \dot{y}, \dot{a}_0, \dot{a}_1)).$$

Lemma 3.5 (The Squaring Argument and Ledger Update). *If the iteration is stuck at stage β , we can replace the factor $\mathbb{P}_{\gamma, \beta}$ with its product square $\mathbb{P}_{\gamma, \beta}^{(2)}$ to neutralize A_m . Furthermore, we can uniquely encode the resulting duplicated Σ_3^1 witnesses into a ledger real R_k .*

Proof. We return to the universe $W[G_\gamma]$ and force with $\mathbb{P}_{\gamma, \beta}^{(2)} \cong (\mathbb{P}(\gamma) * \mathbb{P}_{\gamma+1, \beta}) \times (\mathbb{P}(\gamma) * \mathbb{P}_{\gamma+1, \beta})$. Let $G_{\gamma, \beta}^{(2)}$ be the generic filter.

To apply Lemma 3.3, we must verify the non-interference condition. The restriction sets E^0 and E^1 generated along the two parallel branches of the squared forcing exclusively add restrictions involving their respectively generated Cohen reals. Because mutually generic Cohen reals are strictly distinct, the left branch of the iteration never utilizes a coding forcing for a tuple containing y^1 or a_0^1 , and vice versa. Thus, the non-interference condition is perfectly satisfied, and the product square $\mathbb{P}_{\gamma, \beta}^{(2)}$ is a valid γ -suitable forcing.

The generic $G_{\gamma,\beta}^{(2)}$ yields two mutually generic evaluations of the putative value y , denoted y^0 and y^1 , and provides two evaluations of the name $\dot{a}_0$, denoted a_0^0 and a_0^1 . By the forcing theorem applied in $W[G_\gamma]$, and because the stuck condition quantifies over all suitable forcings (which now mathematically includes product squares and their subsequent suitable extensions), the new ground model $W[G_\gamma][G_{\gamma,\beta}^{(2)}]$ satisfies for each $j \in \{0, 1\}$:

$$\exists a_0^j \forall \mathbb{Q} (\mathbb{Q} \text{ is suitable} \implies \mathbb{Q} \Vdash \forall a_1 \psi_m(x, y^j, a_0^j, a_1)).$$

Because $y^0 \neq y^1$, in any final suitable extension, both (x, y^0) and (x, y^1) will satisfy $\exists a_0 \forall a_1 \psi_m$. Thus, $y^0, y^1 \in (A_m)_x$, ensuring A_m cannot uniformize σ .

Because we squared the interval $[\gamma, \beta)$, any real w whose Σ_3^1 witness was coded into $\vec{S}^0$ during this interval now has *two* potentially distinct witnesses coded along the parallel branches of the product. In $W[G_\gamma][G_{\gamma,\beta}^{(2)}]$, we define a real R_k that explicitly lists all such reals w processed in this interval and their corresponding chosen witnesses.

Because the squared forcing added new mutually generic Cohen reals, $R_k \notin L[R_{k-1}]$ (where $R_{-1} = \emptyset$). Immediately following this squaring step, we append the coding factor $\text{Code}(k, R_k, \emptyset, m^*, 0, \eta)$ to the iteration, locking the pair (k, R_k) into the $\vec{S}^0$ sequence under the reserved index m^* . $\square$

3.4 Bounding the Rollbacks and Π_2^1 -Definability

The squaring operation permanently neutralizes A_m as a potential uniformizing function. Since there are only countably many lightface Σ_4^1 formulas, this rollback and squaring process can occur at most $N \leq \omega$ times. Let $\beta^* < \omega_1$ be the supremum of all stages where a squaring operation occurs.

After stage β^* , the actual iteration $\mathbb{P}_{\omega_1}$ completely stabilizes and runs linearly. If $N < \omega$, we use subsequent odd stages in the linear portion of the iteration to iteratively adjoin fresh Cohen reals R_k for $k \geq N$, coding them sequentially into $\vec{S}^0$ under the index m^* . This ensures the sequence of ledger reals is always of length ω .

Lemma 3.6. *In the final extension $W[G_{\omega_1}]$, the following Π_2^1 formula $\Psi(R^*)$ holds true for exactly one real R^* which pieces together the sequence $(R_k)_{k \in \omega}$:*

$$\begin{aligned} \Psi(R) \equiv & (0, (R)_0, \emptyset, m^*) \text{ is coded into } \vec{S}^0 \wedge \\ & \forall k \in \omega \left((k+1, (R)_{k+1}, \emptyset, m^*) \text{ is coded into } \vec{S}^0 \wedge (R)_{k+1} \notin L[(R)_k] \right). \end{aligned}$$

Proof. By the mechanics of our ledger updates, the true sequence $(R_k)_{k \in \omega}$ generated by the iteration satisfies Ψ . Uniqueness follows from Lemma 2.6 (No Unwanted Codes); because suitable forcings cannot accidentally add codes, any sequence satisfying Ψ must perfectly match the exact ω -sequence of reals explicitly coded under the index m^* by our iteration. Since $(R)_{k+1} \notin L[(R)_k]$ is enforced at every step, the sequence cannot loop or branch. Thus, R^* is uniquely Π_2^1 -definable. $\square$

4 Proof of the Main Theorem

Theorem 4.1. *In the generic extension $W[G_{\omega_1}]$, the lightface Σ_3^1 -uniformization property holds, and the lightface Σ_4^1 -uniformization property fails.*

Proof. We first verify the global failure of lightface Σ_4^1 -uniformization. Consider our designated lightface Σ_4^1 relation $\sigma(v_0, v_1)$ asserting that (v_0, v_1) is not coded into $\tilde{S}^1$. Let A_m be an arbitrary lightface Σ_4^1 relation. Because the iteration stabilized, exactly one of two outcomes occurred for m :

1. **The iteration completed without getting stuck for m :** For every $x \in P_m$, putative values $c_i(x, m)$ were generically added, and their uncoded status was permanently protected by the restriction sets E_β . We successfully forced $(x, c_i(x, m)) \notin A_m$. Thus, $(x, c_i(x, m)) \in \sigma_x \setminus (A_m)_x$.
2. **The iteration got stuck for m :** The Squaring Argument was invoked prior to β^* . The relation A_m was forced to contain both (x, y^0) and (x, y^1) . Because $y^0 \neq y^1$, the relation A_m is not the graph of a function.

In all cases, A_m cannot uniformize σ .

We now verify the truth of lightface Σ_3^1 -uniformization. Let A_m be a lightface Σ_3^1 relation. The raw coding relation for our Σ_3^1 witnesses is:

$$U_m(x, z) \equiv (x, z, \emptyset, m) \text{ is coded into } \tilde{S}^0.$$

Because of the rollbacks that occurred prior to stage β^* , the relation U_m might contain multiple witnesses for reals processed during the squared intervals. However, all such "double codes" are explicitly listed and resolved within the unique ledger real R^* .

We define our uniformizing function $f_m(x)$ entirely lightface by existentially quantifying over the unique R^* :

$$f_m(x) = z \iff \exists R \left(\Psi(R) \wedge \begin{cases} R \text{ specifies } z \text{ as the witness for } x, & \text{if } x \text{ is listed in } R \\ U_m(x, z) \text{ holds,} & \text{if } x \text{ is not listed in } R \end{cases} \right)$$

Because $\Psi(R)$ is a Π_2^1 formula, and the condition inside the large parentheses (reading the continuous real R and checking the Σ_3^1 coding relation U_m) is entirely Σ_3^1 , the conjunction $\exists R(\Pi_2^1 \wedge \Sigma_3^1)$ yields a formula that is strictly Σ_3^1 in complexity without any parameters.

This lightface relation has the exact same domain as A_m , and because it defers to the uniquely defined ledger R for any ambiguity, it is strictly single-valued. Thus, lightface Σ_3^1 -uniformization holds. This completes the proof. $\square$

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